EX: 10 Customize switch with Network modules using CISCO packet Tracer.

Aim:

To customize a modular switch in CISCO packet Tracer by installing Fast Ethernet and gigabit ethernet fiber network modules.

Procedure:

1. CISCO packet Tracer was opened.
2. switch PT empty was selected from network device.
3. Two pcs and one cisco 2911 router were added.
4. The physical tab of the switch was opened.
5. Two PT-switch-NM-1FEE modules → slot 0 & slot 1.
6. switch powered ON
7. The CLI was opened and the switch boot process.
8. PC-0 was connected to Fast ethernet 0/1
9. PC-1 was connected to Fast ethernet 1/1
10. Router is powered on.
11. The router interface gigabit Ethernet 0/0/0
12. The connectivity was tested using the ping command.

commands used :

switch > enable

switch # show ip interface brief

Router > enable

Router # configure terminal

Router (config)# interface gigabit ethernet 0/0/0

Router (config)# ip address 192.168.10.1

255.235.255.0.

Router (config)# exit

Router# write memory

Router # show ip interface brief

Router # ping 192.168.10.2.

Result :

Thus, the modular switch was successfully customized using appropriate network modules in cisco packet tracer.

EX: 1 networking commands.

Aim:

Study of various network commands used in linux and windows.

windows networking commands

1. arp - a :

Displays ARP table (IP ↔ MAC address mapping)

2. hostname :

shows the computer name.

3. ipconfig /all :

Displays complete IP configuration (IP, gateway, DNS, DHCP, MAC)

4. nbtstat - a :

Checks NetBIOS name resolution

5. netstat :

Displays active network connections, routing table, interface statistics.

6. nslookup :

Performs DNS lookup.

7. Pathping :

combines ping + Traceroute with packet loss statistics.

Linux networking commands :

1. ip - shows IP address, network configuration
   ↳ ip address show .

2. ifconfig - to view and configuring networking interfaces.

3. mtr - combines ping + traceroute, shows route, latency and packet loss at each hop.

4. tcpdump - capture and analyse network packets.

5. ping - Test connectivity between systems, measure latency, detect packet loss.

6. nmcli - To configure ethernet connections

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EX: 2 Assigning IP address manually to the computer.

Aim:

To assign an IP address manually to the computer.

Steps to Assign an IP address manually

1. nmcli connection show - view existing connections

2. Set a static IPv4 address

nmcli connection modify "Wired connection 1" \
ipv4.method manual \
ipv4.addresses 192.0.2.1/24 \ - IP address
ipv4.gateway 192.0.2.254 \ - Gateway
ipv4.dns 192.0.2.200 - DNS server.

3. Activate the connection

nmcli connection up "Wired connection 1"

4. Verify IP address

ip address show

5. Verify default gateway

ip route show

6. Test the network

EX: 3 Study of network cables

Aim:

To understand the study of different types of network cables.

Network cables & Ethernet cable making.

Different type of networking cables,

1. unshielded twisted pair (UTP) cable
2. Shielded twisted pair (STP) cable
3. Coaxial cable
4. Fibre optic cable.

UTP:

Cat 3 - 10 Mbps

cat 5 - up to 100 Mbps

cat 5e - 1 Gbps

Advantages: cheap, easy to install, small diameter

Disadvantages: more affected by EMI / noise

uses: Ethernet networks.

STP:

Cat 6/6a - up to 10 Gbps

Advantages: shielded, faster than UTP, less interference.

Disadvantages : Expensive, more installation effort

uses : Gigabit / 10 Gr Ethernet, data centers.

STP :

Category 7 - up to 10 Gbps

Advantages : Better shielding and reduced interference

Disadvantages : Expensive and difficult to install.

Coaxial cable :

Types : RG-6, RG-59, RG-11

Speed : 10 - 100 Mbps

Advantages : High bandwidth, interference resistant, Versatile.

Disadvantages : Limited distance, costly, bulky.

uses : TV networks and High-speed internet connections.

Fibre optic cable :

Types : Single-mode and Multi-mode.

Speed : up to 100 Gbps

Advantages : High speed, High bandwidth, secure, long distance.

Disadvantages : Expensive and requires skilled installation.

making Ethernet straight - through / crossover cable

Tools & materials

category - 5e Ethernet cable

RJ45 connectors x 2

crimping tool

cable stripper

optional cable tester

steps:

Step 1: Prepare the cable

- Put the protective boots/ shields onto the cable.
- strip about 1.5 cm of the outer cable jacket from both ends.

Step 2: untwist the wires

- Separate the 4 twisted pairs
- Arrange the wires according to the required wiring standard.

Step 3: Arrange the wires

straight - through cable:

- use the same wiring order on both ends.
- usually T568B - T568B.

crossover cable :

- use T568A on one end and T568B on the other end.

Step 4: Insert wire into RJ45

- Keep wires straight and in the correct order.
- Trim them evenly.
- Insert them fully into the RJ45 plug.
- Make sure the cable jacket enters the connector.

Step 5: crimp the connector

- Place the RJ45 plug into the crimping tool.
- Press firmly to lock the pins onto the wires.
- Repeat for the other end.

Step 6: complete the cable :

- Repeat the process on the second end.
- For straight cable → same standard on both ends.
- For crossover cable → different standards on each end.

Step 7: Test.

- Connect the cable to a cable tester

or devices.

- check that all connections are working correctly.

Result:

The study of network cables is developed successfully.

EX: 4 LAN setup using switch.

Aim:

To set up and configure a LAN using a switch and Ethernet cables.

1. LAN - Local Area Network.

connects computer / devices within a limited area such as a home, office, or lab.

Allows file sharing, resource sharing, communication, and data transfer.

2. Requirements:

- 4 computers
- 1 switch (8/16/24 ports)
- 4 Ethernet cables

3. Steps to set up LAN:

Step 1: Plan the Network

Design the network topology according to the number of computers.

Step 2: connect computers

connect each computer to the switch using Ethernet cables.

Step 3: configure IP address.

Go to :

Control Panel → Network connections →

Ethernet Properties → IPV4 → Properties

Use static IP addresses :

Computer	IP Address	Subnet Mask
PC 1	10.1.1.1	255.0.0.0
PC 2	10.1.1.2	255.0.0.0
PC 3	10.1.1.3	255.0.0.0
PC 4	10.1.1.4	255.0.0.0

Keep all computers in the same network.

Step 4: Configure the switch.

- Connect the computer to the switch
- Open the switch's web interface
- Log in using the default credentials
- Configure the switch IP:

  IP address : 10.1.1.5

  Subnet Mask : 255.0.0.0.

Step 5: Test connectivity.

- Open command prompt.
- Use the ping command to check communication between computers.

Example:

Ping 10.1.1.2.

step 6: File sharing

- Select a folder → properties → sharing → share with Everyone.
- Access the shared folder from other computers on the same LAN.

step 7: Observation

- Draw the LAN topology diagram.
- Record the IP configuration of each device.
- write the configuration steps and challenges faced.

Result:

The LAN setup using switch was successfully developed.

EX: 5 Design a simple topology and configure with one router, two switches and PCs using CISCO PACKET TRACER

Aim:

To design and configure a simple network topology using one router, two switches, and four PCs in Cisco packet Tracer.

Procedure:

1. Place one router, two switches, and four PCs in Packet Tracer.

2. Connect the PCs to the switches and the switches to the router using straight through cables.

3. Assign IP addresses to all PCs in two different networks.

4. Configure the router interfaces with the corresponding gateway IP addresses.

5. Configure the default gateway on each PC.

6. Use the ping command to test connectivity between PCs.

Observation:

1. All devices were connected successfully

2. Router interfaces became active after using no shutdown.

3. PCs in the same network communicated successfully.

4. PCs in different networks communicated through the router.

5. Ping response confirmed successful connectivity.

Result:

The network was successfully created and configured in Cisco Packet Tracer.

Ex: 6 Building an anonymous FTP scanner using python.

Aim:

To develop a python-based anonymous FTP scanner using the built-in ftplib module and determine whether an FTP server permits anonymous access.

Requirements:

- windows / Linux computer
- Python 3.x
- Python built-in ftplib module
- Authorized FTP server for Testing
- Internet / Network connection.

Procedure:

1. Install python 3.x on the computer.
2. open command prompt or Terminal
3. Check the python installation
4. Create a python file named anonymous-ftp-scanner.py
5. Import the FTP and all_errors functions from the ftplib module

6. Define the anonymous_ftp_scan() function

7. Enter the authorized FTP server address.

8. Connect to the FTP server using port 21.

9. Test anonymous login using the username anonymous.

10. Check whether the FTP server permits anonymous access.

11. Display the accessible directory contents using nlst()

12. Close the FTP connection and display the result.

Commands used:

1. python --version

2. py --version for windows.

2. anonymous_ftp_scanner.py

3. python anonymous_ftp_scanner.py

4. Enter FTP server address: 127.0.0.1

Result:

The anonymous FTP scanner was successfully developed and executed using python.

EX : 7 Implement packet sniffing using RAW SOCKETS in python

Aim :

To implement a packet sniffer using RAW sockets in python and capture network packets by displaying important IP header information such as source IP, destination IP, protocol TTL and packet length.

Requirements :

- windows / linux computer
- Python 3.x
- visual studio code (optional)
- command prompt / terminal
- Administrator.

Procedure :

1. Install python and add it to the system path.
2. verify the python installation using command prompt.
3. Install Visual studio code and the python extension if required.

4. Create a project folder named PacketSniffer

5. Create a python file named sniffer.py

6. Import the socket and struct modules.

7. Create a RAW socket using socket.AF_INET socket.SOCK_RAW, and socket.IPPROTO_IP.

8. Obtain the local machine IP address and bind the socket.

9. Enable IP header inclusion and promiscous mode.

10. Receive raw packets using recvfrom().

11. Extract and decode the IP header using struct.unpack().

12. Display the IP version, Header length, TTL, protocol, source IP, destination IP, and packet length.

13. Run the program using Administrator / root privileges.

14. Stop the packet sniffer using ctrl + c.

commands used :

1. python --version

2. python

3. cd desktop
   cd packetsniffer

4. python sniffer.py

   In Linux

5. sudo python3 sniffer.py

Result :

The packet sniffing program was successfully implemented using RAW sockets in python.

EX: 8 Nmap live Host Discovery.

Aim :

To discover live hosts on a network using Nmap host-discovery scans such as ARP, ICMP, TCP and UDP ping scans, and to understand the protocol used at each layer of the TCP/IP model.

Requirements :

- TryHackMe account with accss to the Nmap Live Host Discovery room.
- TryHackMe Attackbox or Kali/Linux VM
- Basic knowledge of TCP/IP and subnetting
- Linux terminal
- Nmap tool.

ARP - link layer

ICMP - Network layer

TCP - Transport layer

UDP - Transport layer.

Procedure:

1. Log in to TryHackMe and open the Nmap Live Host Discovery room.

2. Start the target network and launch the AttackBox.

3. Open the terminal in the AttackBox.

4. Note the target subnet / connection - ID

5. Perform an ARP host discovery scan using -PR.

6. Perform ICMP echo, Timestamp and address-mask scans using -PE, -PP, -PM

7. perform TCP syn and TCP ACK ping scans using -PS and -PA.

8. Perform UDP ping discovery using -PU

9. Compare the number of live hosts detected by each method.

10. Observe the protocol and layer used by each scanning technique.

Commands used :

1. sudo nmap -PR -sn 10.200.6.0/24

2. sudo nmap -PE -sn 10.200.6.0/24

3. nmap -PP -sn 10.200.6.0/24

4. nmap -PM -sn 10.200.6.0/24

5. nmap -PS -sn 10.200.6.0/24

6. sudo nmap -PA -sn 10.200.6.0/24

7. sudo nmap -PU -sn 10.200.6.0/24.

Result :

The Nmap Live Host Discovery experiment was successfully performed.

EX: 9 CISCO PACKET TRACER
(simulation Tool)

Aim: To study the installation and user interface of cisco packet tracer and understand its environment for designing simple networks.

components:

1. menu Bar:
used for file operations and settings.

2. main Toolbar:
Provides shortcuts such as open, save, zoom, undo and redo.

3. Logical / physical:
switches between logical and physical views.

4. workspace:
Area used to create and view topologies.

5. Realtime / simulation tabs.

6. user-created packet Box.

Procedure:

1. Open cisco packet tracer.
2. Place 4 PCs and 1 Hub in the workspace
3. Place another 4 PCs and 1 switch.

EX: 9 CISCO PACKET TRACER
(simulation Tool)

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   switches between logical and physical views.

4. workspace:
   Area used to create and view topologies.

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Procedure:

1. Open cisco packet trace.
2. Place 4 PCs and 1 Hub in the workspace
3. Place another 4 PCs and 1 switch.

4. Assign IP addresses and subnet masks to the PCs.

5. Repeat the process for PCs connected to the switch.

6. Observe and compare how the Hub and switch forward packets.

Result:

Thus, Cisco packet tracer installation, user interface, and the behaviour of Hub and switch were studied successfully.